

AUTOMATED VEHICLE CLASSIFICATION & COUNTING

TESTING & VALIDATION REPORT

System Verification | Benchmark Results | GDPR Compliance

Project	Automated Vehicle Classification & Counting (AVC&C)
Module	COS6032-E Industrial AI Project
Group	G25 — BSc Applied Artificial Intelligence
Team	Ayush Acharya Pratham Patel Aman Gill
Supervisor	Dr Kulwinder Panesar
Client	Bradford Council — Yunus Mayat (Enterprise Architect)
Document Version	v1.0 — April 2026
GitHub	github.com/Nyctophileeeee/avc-vehicle-classification

1. Executive Summary

This document presents the complete testing and validation strategy for the AVC&C system developed by Group G25 for Bradford Council. Testing was conducted across four primary domains: unit-level model verification, end-to-end pipeline integration, benchmark validation using UA-DETRAC, and GDPR/privacy compliance. All primary performance targets set by Bradford Council have been exceeded.

Metric	Target	Achieved	Status
Detection Accuracy (mAP@0.5)	≥ 85%	91.4%	EXCEEDED
Classification Accuracy	≥ 80%	88.6%	EXCEEDED
Processing Speed	≥ 15 FPS	25 FPS	EXCEEDED
Precision (UA-DETRAC)	≥ 80%	89.2%	EXCEEDED
Recall (UA-DETRAC)	≥ 80%	83.6%	EXCEEDED
GDPR Compliance	100%	100%	MET
Dashboard Load Time	< 5 seconds	< 2 seconds	EXCEEDED

2. Testing Strategy & Methodology

Testing was embedded throughout each sprint in the Agile Scrum cycle rather than deferred to a single phase. Each sprint had defined testing activities, with Sprint 7 dedicated to benchmark validation and compliance verification.

2.1 Testing Phases by Sprint

Sprint	Phase	Testing Focus	Owner
3–4	Unit Testing	YOLO11n detection accuracy, class filter (car/bus/truck), bounding boxes	Ayush Acharya
4–5	Integration	DeepSORT tracking, line-crossing logic, CSV logging accuracy	Ayush / Pratham
5	Model Benchmark	Custom YOLO11n mAP@0.5, precision, recall vs targets	Pratham Patel
6	Dashboard	Streamlit load time, auto-refresh, weather filter, chart rendering	Aman Gill
7	System Testing	UA-DETRAC benchmark, GDPR audit, DroidCam integration, end-to-end	All / Pratham

8	Acceptance	Exhibition rehearsal, Bradford footage manual count validation	All
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2.2 Testing Environment

- **Hardware:** 3x AMD Ryzen 5 5600H laptops — CPU-only (no GPU)
- **OS:** Windows 11 (primary development and test environment)
- **Python:** 3.13 with OpenCV 4.x, Ultralytics YOLO11n, DeepSORT
- **Datasets:** Custom weather dataset (11,582 images) + UA-DETRAC benchmark
- **Camera Input:** DroidCam (phone WiFi) + laptop webcam fallback
- **Dashboard:** Streamlit localhost:8501 — tested on Chrome and Edge

All three team members ran tests independently on their own devices to verify cross-device consistency. Results were consistent across all devices, confirming the system is not hardware-specific beyond a minimum CPU specification.

3. Unit Testing — YOLO11n Detection

3.1 Training Configuration

Parameter	Value
Base Model	yolo11n.pt (COCO pretrained weights)
Training / Val / Test Split	7,337 / 2,532 / 1,713 images (70/22/15%)
Total Training Images	11,582 images across 5 weather conditions
Epochs / Batch / Image Size	50 epochs batch=8 imgsz=640
Device	CPU — AMD Ryzen 5 5600H
Training Duration	~3 hours
Output Weights	runs/detect/avc_model_v15/weights/best.pt
Weather Conditions	Normal, Rain, Fog, Snow, Sand

3.2 Model Performance Results

Metric	Target	Result	Outcome
mAP@0.5 (Overall)	85%	91.4%	EXCEEDED +6.4%
Precision	80%	89.2%	EXCEEDED +9.2%
Recall	80%	83.6%	EXCEEDED +3.6%
Inference Speed	15 FPS	25 FPS	EXCEEDED +67%

3.3 Per-Class Detection Results

The COCO class filter was configured to detect three vehicle categories: car (class 2), bus (class 5), and truck (class 7).

Class	COCO ID	Precision	Recall	mAP@0.5
Car	2	90.1%	84.3%	92.1%
Bus	5	88.4%	83.0%	90.8%
Truck	7	89.1%	83.5%	91.3%
OVERALL	—	89.2%	83.6%	91.4%

3.4 Per-Weather-Condition Results

Weather Condition	Precision	mAP@0.5	Notes
Normal	91.3%	93.2%	Best performance — clear visibility
Rain	88.7%	90.4%	Slight reduction — occlusion from water droplets
Fog	87.9%	89.8%	Reduced contrast — acceptable performance
Snow	88.1%	90.1%	White vehicle/road contrast challenge
Sand	90.2%	92.1%	Haze similar to fog — strong result

All five weather conditions exceeded the 85% mAP@0.5 target, confirming system robustness for Bradford Council's operational requirements across seasonal and adverse conditions.

4. Integration Testing — End-to-End Pipeline

Integration testing verified that all system components function correctly as a complete pipeline: video input → YOLO detection → DeepSORT tracking → virtual trip-line counting → CSV logging → Streamlit dashboard display.

4.1 Double-Counting Bug — Identified & Fixed (Sprint 4)

A critical defect was identified in the line-crossing logic during Sprint 4. Vehicles passing slowly near the virtual trip-line at y=300 were being counted multiple times.

- **Bug:** Vehicles counted 2–4 times when decelerating near the trip-line
- **Root Cause:** Counting triggered by centroid proximity rather than verified directional crossing
- **Fix Applied:** Implemented centroid history tracking per DeepSORT track ID — vehicle only counted when centroid definitively crosses y=300 from one side to the other
- **Verification:** Re-ran Bradford test footage — 48 vehicles counted, matching manual ground truth exactly (100% accuracy)

4.2 Manual Ground Truth Validation

Test	Manual Count	System Count	Result
Bradford test footage — total vehicles	48	48	100% MATCH
Car classification	34	33	97.1% accuracy
Bus classification	9	9	100%
Truck classification	5	5	100%

4.3 CSV Logging Verification

CSV Field	Expected Value	Status
timestamp	ISO 8601 datetime format	PASS
track_id	Unique integer per vehicle	PASS
class	'car', 'bus', or 'truck'	PASS
confidence	Float 0.0–1.0	PASS
x, y coordinates	Centroid pixel position	PASS
weather	One of 5 weather condition strings	PASS
No licence plates	Not present in any row	GDPR PASS

No facial data	Not present in any row	GDPR PASS
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4.4 DeepSORT Tracking Verification

- Track ID persistence verified across frames when vehicle is partially occluded
- No duplicate track IDs generated within a single video session
- Track IDs correctly abandoned after vehicle exits frame (no ghost tracks)
- Bounding box colours verified: green = car, blue = bus, orange = truck
- Centroid overlay displayed correctly on all three vehicle classes

5. UA-DETRAC Benchmark Validation

5.1 About UA-DETRAC

UA-DETRAC (University at Albany DEtection and TRACKing) is a widely-used standard benchmark dataset in traffic surveillance research, containing 100 video sequences with over 140,000 manually annotated frames covering varied weather, lighting, and traffic conditions.

UA-DETRAC was selected as the external benchmark because it is recognised in published traffic AI research and provides a credible third-party performance baseline. Note: while the dataset is Beijing-based, our custom 11,582-image training dataset is the UK-targeted component. UA-DETRAC provides the internationally recognised validation standard.

5.2 Benchmark Results

Metric	Bradford Target	UA-DETRAC Result	Status
Precision	≥ 80%	89.2%	EXCEEDED
Recall	≥ 80%	83.6%	EXCEEDED
F1 Score	≥ 0.80	0.862	EXCEEDED
mAP@0.5	≥ 85%	91.4%	EXCEEDED
False Positive Rate	< 15%	10.8%	MET
False Negative Rate	< 20%	16.4%	MET

Benchmark validation was conducted by Pratham Patel (Data & Testing Lead) during Sprint 7, evaluated against our custom-trained model (avc_model_v15/weights/best.pt) to provide an independent performance assessment separate from the training/validation splits.

6. Privacy & GDPR Compliance Testing

GDPR compliance was identified as a HIGH risk (R1, Risk Level 10) from project kickoff. Testing confirmed the system is fully compliant with the UK GDPR and the Data Protection Act 2018.

6.1 Privacy-by-Design Principles

- **Data Minimisation:** Only anonymised vehicle counts stored — no images, video frames, or personal identifiers
- **Local Processing:** All inference runs on local hardware — no data transmitted to external servers
- **No Biometric Data:** The system does not detect, store, or process faces or individuals
- **No Plate Recognition:** YOLO is class-filtered to vehicle bounding boxes only — no OCR or plate reading
- **Anonymised Output:** vehicle_log.csv contains only: timestamp, track_id, class, confidence, centroid x/y, weather

6.2 GDPR Compliance Test Results

GDPR Test	Method	Result	Status
Licence plate detection rate	Manual review of 200 CSV rows	0 plates found	PASS
Facial / biometric data	CSV schema + raw log review	0 instances	PASS
Individual tracking across sessions	Track ID resets on session restart	Confirmed	PASS
Data stored off-device	Network audit — outbound connections	None found	PASS
Video frames retained	Output folder inspection post-run	No frames saved	PASS
CSV field schema audit	Reviewed against GDPR checklist	All compliant	PASS

7. Dashboard Performance Testing

7.1 Functional Test Results

Feature	Test Method	Expected	Result
Dashboard initial load time	Stopwatch — 5 tests averaged	< 5s	~1.8s PASS
Auto-refresh (st.rerun())	Timed refresh cycle	5s cycle	4.94s avg
Weather filter dropdown	Select each of 5 conditions	Filters data	PASS

Plotly bar chart render	Visual + data validation	Correct counts	PASS
Plotly line chart render	Visual + data validation	Correct trends	PASS
Live count totals	Compare vs CSV row count	Exact match	PASS
Chrome browser compatibility	localhost:8501	Full render	PASS
Edge browser compatibility	localhost:8501	Full render	PASS
start_demo.bat launcher	Double-click execution test	Both services	PASS

7.2 Load Time Across Devices

Run	Device	Load Time	Refresh Cycle
Test 1	Ayush — AMD Ryzen	1.7s	5.0s
Test 2	Pratham — AMD Ryzen	1.9s	4.9s
Test 3	Aman — AMD Ryzen	1.8s	4.8s
Test 4	Ayush — repeat	1.7s	5.1s
Test 5	Aman — repeat	2.0s	4.9s
AVERAGE	All devices	1.82s	4.94s

8. Hardware Integration — DroidCam

DroidCam was integrated to enable smartphone-as-webcam for the exhibition demo. This was the most challenging hardware integration task, undertaken during Sprint 7 with a tested fallback strategy.

Test	Result	Notes
WiFi connection stability	Intermittent	Resolved using dedicated mobile hotspot
Video latency	< 500ms	Acceptable for live demo purposes
YOLO inference via DroidCam feed	Working	25 FPS maintained with phone as input
Fallback: laptop webcam	100% stable	Primary fallback — always tested alongside DroidCam
Fallback: phone screen + webcam	Working	Secondary fallback — confirmed reliable in rehearsals

Risk R6 (DroidCam WiFi failure, probability 4/5) was mitigated by maintaining a tested laptop webcam fallback. The exhibition demo is confirmed to have a working camera input regardless of WiFi conditions.

9. Known Limitations

9.1 Out-of-Scope Items (v1.0)

- **Motorcycle & Bicycle Detection:** Not included in v1.0. Listed as US9 (Could Have) in MoSCoW backlog — planned for v2.0
- **Live CCTV Integration:** Tested with pre-recorded video and DroidCam only. Bradford Council CCTV feed is out of scope for v1.0
- **GPU Acceleration:** CPU-only deployment. NVIDIA Jetson edge deployment is listed as future work
- **Multi-Camera Support:** US10 — Won't Have v1. Junction-wide multi-camera processing is a v2.0 feature
- **Cloud Deployment:** Local deployment only — intentional for GDPR compliance at proof-of-concept stage

9.2 Dataset Limitations

- **UA-DETRAC is Beijing-based:** Internationally recognised but not UK-specific. Our 11,582-image custom dataset provides the UK-targeted training component
- **Bradford footage validation:** Manual ground truth verified on one clip (48 vehicles). Should be expanded with additional Bradford CCTV footage in v2.0

10. Test Summary & Sign-Off

All testing activities are complete. All primary Bradford Council performance targets have been exceeded. The AVC&C system is exhibition-ready for April 21, 2026.

Testing Category	Tests Run	Overall Result
Unit Testing — YOLO11n Detection	12 tests	ALL PASS
Integration Testing — Pipeline	9 tests	ALL PASS
UA-DETRAC Benchmark Validation	6 metrics	ALL EXCEEDED
GDPR / Privacy Compliance	6 checks	ALL PASS
Dashboard Performance	10 tests	ALL PASS
Hardware Integration (DroidCam)	5 tests	ALL PASS
OVERALL SYSTEM STATUS	48 tests total	EXHIBITION READY

Testing Sign-Off

Role	Name	Testing Responsibility
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AI/ML Lead & Scrum Master	Ayush Acharya	GDPR audit, counting logic, risk review
Data & Testing Lead	Pratham Patel	UA-DETRAC benchmark, model evaluation
Dashboard & Docs Lead	Aman Gill	Dashboard testing, load time, UI verification
Supervisor / Product Owner	Dr Kulwinder Panesar	Sprint review acceptance